

MUAN INTL, Korea, Republic of

WMO: 471630

Lat: 34.983N

Lon: 126.383E

Elev: 16

StdP: 101.13

Time Zone: 9.00 (E09)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.1	(c) -4.9	(d) -11.8	(e) 1.4	(f) -2.2	(g) -10.0	(h) 1.6	(i) -2.6	(j) 11.0	(k) -1.0	(l) 10.3	(m) -0.8	(n) 3.7	(o) 50	(p) 0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.4	(c) 32.8	(d) 26.4	(e) 31.2	(f) 26.1	(g) 30.1	(h) 25.5	(i) 27.4	(j) 30.0	(k) 26.9	(l) 29.6	(m) 26.3	(n) 28.9	(o) 4.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	28.6	26.1	21.6	28.2	25.8	21.1	28.1	86.8	30.5	84.9	29.3	82.2	28.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.9	7.9	-8.5	34.7	2.3	0.9	-10.2	35.4	-11.5	35.9	-12.8	36.4	-14.5	37.1		
			-9.3	28.2	2.2	1.2	-10.9	29.1	-12.2	29.8	-13.4	30.5	-15.0	31.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	1.0	2.8	6.9	12.5	18.0	22.3	26.2	26.9	22.4	16.5	10.1	3.6		
	DBStd	9.34	3.10	3.36	3.28	3.03	2.40	1.66	2.10	2.22	2.28	2.81	3.79	3.46		
	HDD10.0	849	278	203	109	12	0	0	0	0	1	45	201			
	HDD18.3	2316	536	435	356	177	36	0	0	0	1	70	247	458		
	CDD10.0	2367	0	2	12	86	248	369	502	524	373	201	49	2		
	CDD18.3	792	0	0	0	1	26	119	244	266	124	12	1	0		
	CDH23.3	5962	0	0	0	3	127	468	2097	2591	637	37	1	0		
	CDH26.7	1664	0	0	0	0	9	46	607	899	102	2	0	0		
(14) Wind	WSAvg	3.5	3.8	4.0	4.0	3.7	3.3	3.0	3.4	3.2	3.0	3.3	3.6	3.9		
(15) Precipitation	PrecAvg	1228	31	42	69	87	99	155	237	229	134	53	55	37		
	PrecMax	1754	67	106	145	211	246	381	496	463	450	199	154	81		
	PrecMin	682	8	9	22	11	30	20	26	24	17	4	6	5		
	PrecStd	259	15	24	36	51	47	85	100	118	109	45	38	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	17.0	20.0	23.8	27.9	29.1	33.1	34.2	30.2	26.1	21.9	15.9		
		MCWB	8.8	13.8	14.8	15.5	18.4	21.3	26.5	26.3	23.4	20.4	17.0	12.8		
	2%	DB	9.2	13.2	17.0	21.9	26.0	28.0	32.0	33.1	29.0	24.1	19.2	12.8		
		MCWB	6.4	10.1	12.9	14.6	18.1	21.4	26.2	26.5	23.3	18.8	15.4	9.9		
	5%	DB	7.8	10.8	15.0	20.0	24.9	27.0	31.0	32.1	27.9	22.9	17.8	10.8		
		MCWB	5.5	7.6	11.5	13.7	17.9	21.1	26.1	26.2	22.3	17.8	15.0	8.4		
	10%	DB	6.1	8.2	13.0	18.1	23.2	26.0	30.0	31.0	26.8	21.2	16.1	9.1		
		MCWB	3.7	5.9	10.0	13.1	17.5	20.9	25.7	26.2	22.1	17.1	13.5	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	14.5	15.7	18.1	22.3	24.5	28.2	28.2	25.8	23.0	18.2	13.7		
		MCDB	11.0	16.0	17.5	21.3	23.9	26.4	30.1	31.1	28.1	25.2	20.9	14.9		
	2%	WB	7.3	11.0	13.8	16.5	20.2	23.6	27.4	27.5	24.7	21.1	16.7	10.5		
		MCDB	8.8	12.4	16.2	19.5	23.3	25.5	29.6	30.4	27.0	22.5	18.3	12.2		
	5%	WB	5.4	8.0	12.2	15.3	19.4	22.9	26.9	27.0	23.9	19.5	15.4	8.6		
		MCDB	7.0	9.9	14.6	18.2	22.5	24.8	29.3	30.0	26.0	21.1	17.1	10.4		
	10%	WB	4.0	6.0	10.4	14.1	18.5	22.2	26.2	26.6	23.0	18.1	13.9	7.0		
		MCDB	5.8	7.8	12.6	17.0	21.8	24.3	28.6	29.4	25.5	20.7	16.1	8.7		
(35) Mean Daily Temperature Range	MDBR		7.3	7.7	8.9	9.5	9.4	7.1	5.8	6.4	7.6	8.8	8.2	7.4		
	5% DB	MCDBR	9.8	11.2	12.0	12.8	11.6	8.7	7.2	8.1	8.8	10.1	9.3	9.6		
		MCWBR	7.3	8.5	8.3	6.7	5.3	3.8	3.1	3.0	4.0	5.5	6.3	7.5		
	5% WB	MCDBR	8.9	9.9	10.5	9.3	8.9	6.0	6.2	7.0	6.7	6.8	7.8	8.6		
		MCWBR	7.3	8.2	8.0	6.1	5.2	3.4	3.3	3.1	3.8	5.1	6.3	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.457	0.497	0.563	0.577	0.595	0.657	0.615	0.562	0.505	0.480	0.468	0.456		
	taud		1.932	1.866	1.736	1.754	1.769	1.717	1.864	2.002	2.089	2.016	1.980	1.943		
	Ebn at Noon		700	722	712	731	725	679	705	736	758	732	684	668		
	Edh at Noon		148	176	218	224	224	235	202	173	151	149	140	138		
(44) All-Sky Solar Radiation	RadAvg		2.33	3.15	4.35	5.06	5.50	4.92	4.63	4.79	4.25	3.72	2.57	2.06		
	RadStd		0.19	0.29	0.39	0.40	0.60	0.46	0.62	0.66	0.41	0.32	0.27	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+59	N/A

Nomenclature: See separate page